Ogbonna Arinze

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SUMMARY

Senior Full Stack Engineer with 8 years of professional experience building high-impact web applications and internal data platforms across healthcare, fintech, civic tech, and mobility sectors. Proven expertise in architecting React and TypeScript frontends with strong backend capabilities in NodeJS, NestJS, PostgreSQL, and REST API development. Experienced in building intuitive internal tooling for technical and non-technical users, optimizing high-traffic systems, and delivering data visualization dashboards that drive business decisions. Adept at cross-functional collaboration, performance optimization, and driving engineering excellence through automated testing and CI/CD pipelines.

CORE COMPETENCIES

- **Full Stack Development:** React, Next.js, TypeScript, NodeJS, NestJS, REST APIs, SQL/PostgreSQL, MongoDB, and Prisma ORM.
- **Frontend Architecture:** Scalable component systems, design systems, monorepos (Nx, Turborepo), SSR/SSG, Redux, Zustand, and Tailwind CSS.
- **Data Visualization & Dashboards:** Recharts, react-chartjs, dynamic charting, real-time data streaming (SSE, EventSource API), and interactive UIs.
- **Backend & Data:** NestJS, NodeJS REST APIs, PostgreSQL, MongoDB, Prisma, GraphQL, JWT authentication, RBAC, and database schema design.
- **Testing & Quality:** Playwright, Jest, Cypress, Allure Reports, Storybook, ESLint, Prettier, and CI-driven QA.
- **Cloud & Infrastructure:** AWS (CloudFront, S3), Docker, GitHub Actions, CI/CD pipelines, Vercel, and Nx monorepo orchestration.
- **Performance & Observability:** Core Web Vitals, code splitting, CDN strategies, Lighthouse, Sentry, Datadog RUM, and bundle analysis.
- **Leadership:** Technical direction, architecture reviews, mentorship, Agile/Scrum, cross-functional collaboration, and product-minded development.

WORK EXPERIENCE

Genentech (Roche) — San Francisco, CA (Remote)

Frontend Engineer — *May 2023 - Present*

- Spearheaded the redesign of the main clinical trial data product, migrating from a **Vue.js** web application to a **React Chrome extension**, optimizing workflows for clinical site integrations and enhancing adoption across internal teams.
- Architected a **reusable design system** in **React** and **Radix UI** with **Tailwind CSS**, improving UI consistency, accessibility (WCAG), and developer velocity across multiple product teams.
- Co-developed a full-stack **Next.js** application powered by a **RAG system** using **OpenAI GPT models**, **LangChain**, **Vercel AI SDK**, and **Chroma** vector database, enabling users to query data in real time via a conversational interface.
- Optimized application performance by implementing **lazy loading, code splitting, dynamic imports**, and **CDN asset delivery**, reducing load times significantly as measured via **Lighthouse** and Chrome DevTools.
- Implemented **JWT-based authentication** and **role-based access control** in Next.js with **TypeScript** and **React Query**, using HTTP-only cookies for secure token handling and permission-based UI rendering.
- Introduced **Redux** for state management and **React Query** for request caching, refactoring legacy class components to functional components and streamlining data flow across the application.
- Increased release reliability by integrating **Playwright end-to-end tests** into the **CI/CD pipeline**, automating reporting with **Allure Reports** on GitHub Pages and configuring Slack notifications for build quality visibility.
- Mentored a team of two junior developers (interns), fostering a culture of technical excellence and code quality.

Descasio — Lagos, Nigeria (Hybrid)

Senior Frontend Developer — *May 2021 - Mar 2023*

- Led the design and implementation of an intuitive **React** user interface, achieving a **25% boost in user engagement** and collaborating with architects to create a scalable frontend architecture that accelerated development by **30%**.
- Optimized frontend performance by profiling render bottlenecks with **React Profiler** and **Lighthouse**, reducing bundle size by **30-40%** through rollup-plugin-visualizer and webpack-bundle-analyzer, achieving a **35% faster load time**.
- Drove a **40% improvement** in application performance by optimizing caching mechanisms across the React application.
- Enhanced development workflow by introducing **Jest** and **Enzyme** testing frameworks, leading to a **20% decrease** in bug reports.
- Utilized **Storybook** for documenting React components, streamlining change management and supporting agile development practices.

- Established engineering standards using **ESLint, Prettier, Husky, Lint-staged**, and **GitHub Actions**, reducing errors and accelerating CI/CD processes.
- Mentored junior engineers on frontend best practices, fostering high-quality, maintainable code across projects.

ActivEdge Technologies Lagos, Nigeria
Senior React Developer *Oct 2019 - Sep 2021*

- Built a high-volume **bank Settlement Solution** in a team of 3, processing thousands to millions of daily transactions and boosting settlement rates by **80%**, demonstrating experience with high-throughput data systems.
- Developed critical payment modules (MasterCard, VISA, Express-cash, E-commerce) using **React**, **Redux**, **RxJS** for real-time data streaming, and **Axios** for asynchronous API requests.
- Built a real-time account reconciliation solution, boosting daily reconciliation by **60%** through innovative transaction processing and data management strategies.
- Adopted the native **EventSource API** for real-time data streaming, replacing socket.io for faster updates and a more responsive interface.
- Built dynamic **data visualization dashboards** using **React** and **react-chartjs**, streamlining management of user roles and permissions with **Redux Toolkit** and **Ant Design** UI.

Founders Lair Beijing, China (Remote)
Full Stack Engineer (Contract) *Oct 2020 - Feb 2021*

- Developed **NodeJS RESTful APIs** and translated business requirements into efficient **MongoDB** database models and schemas, ensuring technical solutions met business needs.
- Transformed Figma designs into reusable **React** components and engineered a **CI/CD pipeline** for automated deployment to **Heroku** and **AWS**.
- Mentored two interns through frontend and backend projects, providing technical guidance and task management.

SureTrade Business Solutions Nigeria
Frontend Developer *Mar 2018 - Oct 2019*

- Designed **MongoDB** database schemas and implemented server-side logic in **NodeJS**, building and testing **REST API** endpoints for multiple client-facing applications.
- Developed **SureEdu**, a school management portal in a team of 5, handling staff payroll and attendance tracking, student records (exams, reports, attendance), and guardian communications.
- Responsible for full-stack integration, connecting the **JavaScript** frontend to **NodeJS** backend services, developing user use cases, and converting business requirements into database schemas and models.
- Devised comprehensive documentation for each application, detailing functions, capabilities, and features. Tested user-defined functions, triggers, and API endpoints to ensure reliability.

PROJECTS

OurNigeria ournigeria.ng
Senior Engineer *Feb 2026 - Present*

AI-powered Nigerian budget analysis and corruption tracking platform. 700+ budget documents, 708K+ vector embeddings, 37 states.

- Architected the entire platform as an **Nx monorepo** managing **6 applications**: **NestJS** API backend, **Next.js** citizen-facing web app with **22 interactive chart types** (Recharts), Next.js admin dashboard with usage analytics and ingestion tracking, NestJS document ingestion pipeline, Next.js marketing landing page, and a Remotion video generation service.
- Designed the **PostgreSQL 16** database schema with **Prisma v7**, including pgvector extension for vector storage, and wrote custom migration tooling that filters out framework-managed tables from Prisma's drift detection.
- Built a multi-agent AI system using **Mastra** with 5 specialized agents and a three-stage intent classification pipeline, powered by a hybrid RAG pipeline combining **pgvector** + BM25 full-text search with Reciprocal Rank Fusion and **Cohere** reranking.
- Implemented **SSE streaming** for real-time responses with 15s keepalive heartbeats, token-budgeted context management, and conversation summarization for efficient data handling.
- Configured CI/CD with **GitHub Actions** for automated builds, **Playwright** E2E tests across 4 projects, and **Docker Compose** production deployment with multi-stage Dockerfiles, health checks, and memory limits.

EDUCATION

University of Nigeria Nsukka, Enugu
BSc Computer Science - 4.1 CGPA *Graduation Date: Sep 2019*